
SENTIMENT ANALYSIS SERVICE

So tay ky thuat va bao caophan quyensohuu

Tai lieu handover, readiness review va phan cong ownership

Tech Lead	Le Quang Tuyen
ML Engineering Lead	Duong Binh An
AI Engineering Lead	Do Quoc Trung
Solution Engineering Lead	Duong Hong Quan
Backend Engineering Lead	Pham Duc Long
UI/UX Engineering Lead	Nguyen Le Hong Nhi

Nguyen tac nguon su that: tai lieu nay uu tien implementation trong repository, sau do den cau hinh runtime, tests, va cuoi cung moi den cac bao cao cu. Neu code va tai lieu mau thuan, van hanh se tin code.

Contents

1 Danh gia dieu hanh

1.1 Muc dich kinh doanh

San pham bien feedback van ban thanh tin hieu van hanh co cau truc: sentiment tong the, sentiment theo aspect, dau hieu sarcasm, va giai thich muc token. Gia tri kinh doanh nam o phan tich review, theo doi chat luong dich vu, ho tro triage support, va demo explainable AI.

1.2 Muc dich san pham

Repository dang dong thoi chua hai san pham:

- dich vu inference online qua Angular va FastAPI
- he thong ML offline cho preprocessing, finetuning, evaluation, MLflow, va ONNX export

1.3 Muc dich ky thuat

Codebase the hien duoc day du chuoi phan phoi ML:

- DVC quan ly data flow
- Hugging Face va PEFT cho training
- ONNX runtime cho serving
- Prometheus va Grafana cho observability
- Docker Compose cho dong goi va chay he thong

1.4 Gia tri cot loi

Gia tri chinh khong nam o model moi la. Gia tri nam o viec dong goi sentiment inference thanh mot dich vu co explainability, batch processing, monitoring, va data/model lineage ro rang.

1.5 Danh gia do truong thanh

Moc	Danh gia
Prototype	Da vuot qua. Repo co API contracts, tests, monitoring, Docker, DVC, MLflow, ONNX export.
MVP	Da vuot qua. Co predict, explain, health, metrics va batch CSV.
Advanced MVP	Trang thai hien tai. He thong da co hanh vi san pham end-to-end va vong doi ML khong don gian.
Pre-Production	Moi dat mot phan. Van con thieu auth, CORS dang mo, state trong memory, batch status mock, va chua co durable job orchestration.
Production Ready	Chua dat. Phu hop cho demo, noi bo, va pilot kiem soat chat; chua phu hop enterprise production.

1.6 Vi sao chua san sang production

- `src/main.py` mo toan bo endpoint inference va evaluation ma khong co auth.
- `/batch_status/{job_id}` la mock, khong co queue hay persistent state.
- frontend cho phép chọn ngôn ngữ nhưng `SentimentAnalysisService.predict()` bỏ qua trường lang.
- Grafana được khởi tạo với mật khẩu mặc định admin.
- health check chỉ xác nhận model đã load, không xác minh toàn bộ dependency xuống dưới.

2 Mô hình tinh thần hệ thống

2.1 Business view

```

1 flowchart LR
2   Customer[Khách hàng hoặc Analyst]
3   UI[Angular UI]
4   API[FastAPI Service]
5   Insight[Sentiment và Aspect Insight]
6   Ops[Monitoring và Reports]
7   Improve[Retraining và Export]
8   Customer --> UI --> API --> Insight --> Ops --> Improve

```

2.2 Product view

```

1 flowchart TB
2   A[Single Predict]
3   B[Explain SHAP]
4   C[Batch CSV Predict]
5   D[Background Evaluate]
6   E[Dashboard và Metrics]
7   UI[Frontend]
8   UI --> A
9   UI --> B
10  UI --> C
11  API[Backend]
12  A --> API
13  B --> API
14  C --> API
15  D --> API
16  API --> E

```

2.3 Engineering view

```

1 flowchart LR
2   FE[app/sentiment-analysis-chatbot]
3   BE[src/main.py]

```

```

4  MODEL[src/model]
5  DATA[src/data]
6  TRAIN[src/scripts + src/training]
7  OBS[infra + src/monitoring]
8  FE --> BE --> MODEL
9  DATA --> TRAIN
10 TRAIN --> MODEL
11 BE --> OBS

```

2.4 Infrastructure view

```

1  flowchart LR
2    Browser --> Nginx[Frontend container :80]
3    Nginx --> FastAPI[fastapi_app :8000]
4    FastAPI --> ML[Model runtime in-process]
5    FastAPI --> Metrics[/metrics]
6    Metrics --> Prometheus[:9091]
7    Prometheus --> Grafana[:3000]
8    FastAPI --> Eval[/evaluate]
9    Eval --> MLflow[:5005]

```

2.5 AI/ML view

```

1  flowchart LR
2    Raw[data/raw]
3    Prep[src.data.pipeline.py]
4    Validate[src.data.validators.py]
5    Train[src.scripts.run_finetuning.py]
6    Eval[src.scripts.evaluate_finetuned.py va src.model.evaluate.py]
7    Export[src.scripts.export_onnx.py]
8    Serve[src.model.baseline.py]
9    Raw --> Prep --> Validate --> Train --> Eval --> Export --> Serve

```

3 Luong nghiệp vụ end-to-end

Buoc	Muc tieu kinh doanh	Implementation ky thuat	Owner
Customer	Gui text de duoc phan tich	Nguoi dung tuong tac voi chat input hoac CSV upload trong Angular components.	UI/UX Lead
UI	Thu thap text, voice transcription, hoac CSV	ChatInputComponent phat su kien text; BatchUploadComponent upload file multipart.	UI/UX Lead
API	Validate request va route dung execution path	src/main.py dung Pydantic schema trong contracts/schemas.py.	Backend Lead
Inference	Sinh sentiment, aspects, sarcasm, hoac SHAP	BaselineModelInference goi HF hoac ONNX runtime va zero-shot ABSA.	AI Lead
Result	Tra response co cau truc cho UI hoac caller	PredictResponse, ExplainResponse, va batch response.	Backend Lead

Buoc	Muc tieu kinh doanh	Implementation ky thuat	Owner
Monitoring	Ghi nhan request count va latency	middleware src/monitoring/metrics.py va endpoint /metrics.	trong Solution Lead
Feedback	Xem metrics va report evaluation	Grafana, MLflow, JSON reports reports/ va data/reports/.	trong Solution Lead / ML Lead
Retraining	Nang cap model/adapters	src/scripts/run_finetuning.py, task configs, class weights, LoRA adapters.	ML Lead
Deployment	Dong goi artifact moi de phuc vu runtime	Dockerfiles, Compose, ONNX export scripts, mounted model dirs.	Solution Lead / Backend Lead

4 Kien truc chi tiet

4.1 Frontend

Trach nhien: message flow, voice capture, luu lich su, explain trigger, batch upload, theme control.

Phu thuoc: Angular, HttpClient, local storage, backend /api/*.

Rui ro: contract drift, bug ngon ngu, frontend test it.

Ownership: UI/UX Lead primary, Backend Lead secondary.

4.2 Backend

Trach nhien: startup lifecycle, dependency injection, validation, exception mapping, health, metrics, evaluate trigger, batch orchestration.

Phu thuoc: FastAPI, pandas, model interface, monitoring middleware.

Rui ro: single-process bottleneck, route ton tai nhung chi la mock, khong auth.

Ownership: Backend Lead primary, Solution Lead secondary.

4.3 Inference layer

Trach nhien: sentiment scoring, sarcasm adapter, ABSA extraction, SHAP, ONNX load, language validation.

Phu thuoc: Hugging Face, PEFT, SHAP, onnxruntime, tokenizer va model artifacts.

Rui ro: startup latency, SHAP ton tai chi phi cao, artifact mismatch.

Ownership: AI Lead primary, ML Lead secondary.

4.4 Training layer

Trach nhien: load dataset, dedup, split, xu ly imbalance, LoRA config, trainer, MLflow logging.

Phu thuoc: Datasets, Transformers, PEFT, sklearn, task configs.

Rui ro: data skew, phu thuoc file raw, reproducibility chi dung trong dung environment.

Ownership: ML Lead primary, AI Lead secondary.

4.5 Monitoring layer

Trach nhien: request metrics, dashboard provisioning, alert rules.

Phu thuoc: Prometheus, Grafana, middleware instrumentation.

Rui ro: khong co tracing, chua co business SLI manh.

Ownership: Solution Lead primary, Backend Lead secondary.

4.6 Deployment layer

Trach nhien: image build, service topology, ports, resource limits, network, mounted volumes.

Phu thuoc: Dockerfile, Dockerfile.train, Compose, model va data dirs.

Rui ro: secret handling yeu, default credentials, khong co rollback automation.

Ownership: Solution Lead primary, Backend Lead secondary.

5 Ban do ownership repository

Folder	Muc dich	Primary	Secondary	Business Im- pact	Modification Risk	Knowledge Risk
app/	Angular UI cho nguoi dung	UI/UX	Backend	High	Medium	Medium
contracts/	API va model contracts dung chung	Backend	AI	High	High	Medium
data/	Raw, processed, eval, report artifacts	ML	AI	High	High	High
docs/	Bao cao cu va hand-book	Tech Lead	All leads	Medium	Low	Low
infra/	Prometheus va Grafana provisioning	Solution	Backend	High	Medium	Medium
mlruns/	Local MLflow store	ML	Solution	Medium	Low	Medium
models/	Adapters va ONNX artifacts	AI	ML	High	High	High
notebooks/	Colab/manual pipeline	ML	AI	Medium	Medium	High
reports/	Evaluation va benchmark outputs	ML	AI	Medium	Low	Medium
src/data/	Download, preprocess, validate	ML	AI	High	High	Medium
src/model/	Runtime inference va export logic	AI	ML	High	High	High
src/monitoring/	Prometheus instrumentation	Solution	Backend	Medium	Low	Low
src/scripts/	CLI entrypoints	ML	AI	High	Medium	Medium
src/training/	Training internals	ML	AI	High	Medium	Medium
tests/	Regression safety net	Tech Lead	Owning team	High	Medium	Medium

Folder	Mục đích	Primary	Secondary	Business Impact	Modification Risk	Knowledge Risk
.github/	CI behavior	Backend	Tech Lead	High	Medium	Medium

6 Trace thuc thi runtime

Flow	File / Class	Function	Input	Output	Exception Path	Owner
Startup	src/main.py / module	lifespan()	env MODEL_MODE	ml_model san sang	ModelError khi load loi	Backend + AI
Health	src/main.py / API	health_check()	HTTP GET	HealthResponse	503 neu model chua ready	Backend
Predict	src/main.py / API	predict()	PredictRequest	PredictResponse	400/500 neu lang hoac model loi	Backend
Predict core	src/model/baseline / BaselineModelInference	predict_single()	text, lang	PredictionResponse	unsupported language hoac model failure	AI
Explain	src/main.py / API	explain()	ExplainRequest	ExplainResponse	explainer failure	Backend + AI
Batch Predict	src/main.py / API	batch_predict()	uploaded CSV	BatchPredictionResponse	400 invalid CSV, 500 batch failure	Backend
Batch core	src/model/baseline / BaselineModelInference	predict_batch()	list text	list PredictionResponse	invalid batch size	AI
Evaluate trigger	src/main.py / API	run_evaluate()	POST	status JSON	409 neu dang chay, state memory co the mat	Backend + ML
Evaluate status	src/main.py / API	evaluate_status()	GET	state JSON	stale sau restart	Backend
Offline eval	src/model/evaluate.py	evaluate_on_data_frame	processed test frame	metrics dict	tra ve error payload neu split rong	ML + AI

6.1 Sequence diagrams

Startup

```

1 sequenceDiagram
2     participant U as Uvicorn/FastAPI
3     participant A as src.main.lifespan

```



```

4 participant C as ModelConfig
5 participant M as BaselineModelInference
6 participant O as OnnxInferenceSession hoac HF model
7 U->>A: app startup
8 A->>C: doc MODEL_MODE
9 A->>M: khoi tao
10 M->>O: load tokenizer/model/session
11 A->>M: preload()
12 M->>M: warm ABSA pipeline
13 A-->>U: app ready

```

Predict

```

1 sequenceDiagram
2 participant User
3 participant FE as Angular UI
4 participant API as FastAPI /predict
5 participant LD as LanguageDetector
6 participant MI as BaselineModelInference
7 participant ABSA as Zero-shot pipeline
8 User->>FE: nhap text
9 FE->>API: POST /api/predict {text}
10 API->>LD: resolve_request_language()
11 API->>MI: predict_single(text, lang)
12 MI->>MI: sentiment logits
13 MI->>ABSA: aspect extraction
14 MI-->>API: PredictionResult
15 API-->>FE: PredictResponse
16 FE-->>User: render ket qua

```

7 Team handbooks

7.1 7.1 Tech Lead Handbook

Owner: Le Quang Tuyen.

Responsibilities: kien truc, code-review bar, readiness approval, giai quyot xung dot cross-team.

Systems owned: kien truc toan he thong, standards, quality gates.

Operational duties: review readiness va chi huy incident Sev-1.

Release duties: approve contract changes va security exceptions.

Risks: architecture drift va ownership mo ho.

KPIs: escaped defects, release predictability, readiness score.

7.2 7.2 ML Engineering Handbook

Owner: Duong Binh An.

Responsibilities: datasets, preprocessing, validation thresholds, finetuning, evaluation, MLflow.

Systems owned: src/data/, src/training/, scripts training, data/, reports/.

Operational duties: dam bao dataset integrity va evaluation reproducibility.

Release duties: xac nhan metric truoc khi promote artifact.

Risks: knowledge silo o data lineage va imbalance logic.

KPIs: macro F1, per-language F1, validation pass rate.

7.3 7.3 AI Engineering Handbook

Owner: Do Quoc Trung.

Responsibilities: inference behavior, ONNX runtime, ABSA, sarcasm, SHAP, language detection.

Systems owned: `src/model/`, `models/`.

Operational duties: model load diagnostics, latency tuning, artifact compatibility.

Release duties: ky duyệt artifact integrity va inference regression.

Risks: artifact mismatch, SHAP chi phi cao, provider issues.

KPIs: p95 latency, load success, inference error rate.

7.4 7.4 Solution Engineering Handbook

Owner: Duong Hong Quan.

Responsibilities: integration end-to-end, monitoring stack, deployment topology.

Systems owned: `infra/`, Compose topology, release runbooks.

Operational duties: giu observability va environment consistency.

Release duties: verify service health, ports, volume, dashboards.

Risks: dependency drift, insecure defaults, nhieu buoc manual.

KPIs: deployment success, monitoring freshness, MTTR.

7.5 7.5 Backend Engineering Handbook

Owner: Pham Duc Long.

Responsibilities: API semantics, validation, exception handling, evaluate va batch endpoints.

Systems owned: `src/main.py`, `contracts/`, backend CI.

Operational duties: endpoint health, error triage, backward compatibility.

Release duties: giu OpenAPI dung va schema on dinh.

Risks: route drift, state memory, expensive call khong gioi han.

KPIs: 5xx rate, schema-change defect rate.

7.6 7.6 UI/UX Engineering Handbook

Owner: Nguyen Le Hong Nhi.

Responsibilities: user journey, interaction quality, explain visualization, batch UX.

Systems owned: Angular application.

Operational duties: validate client behavior voi backend contract.

Release duties: dam bao feature hien thi dung voi backend support that.

Risks: contract mismatch, test it, state local storage.

KPIs: task completion, UI defect rate, explain usage.

8 RACI matrix

Workstream	Tech	ML	AI	Solution	Backend	UI/UX
Feature development	A	C	C	C	R	R
Model training	C	A/R	C	I	I	I
Model deployment	A	C	R	R	C	I
API changes	A	I	C	C	R	C
Production release	A	C	C	R	R	C
Incident management	A	C	R	R	R	I
Monitoring	C	I	C	A/R	R	I
Security	A	I	C	R	R	C
Architecture changes	A/R	C	C	C	C	C

9 Data va model lineage

```

1 flowchart LR
2   Raw[data/raw/*.csv]
3   Validation1[src.data.downloader.validate_raw_schema]
4   Prep[src.data.pipeline.PreprocessingPipeline.run]
5   Validation2[src.data.validators.DataQualityValidator.validate]
6   Train[src.scripts.run_finetuning.train]
7   Eval[src.model.evaluate hoac src.scripts.evaluate_finetuned]
8   Track[MLflow runs]
9   Export[src.scripts.export_onnx]
10  Runtime[src.main -> BaselineModelInference]
11  Predict[PredictResponse]
12  Raw --> Validation1 --> Prep --> Validation2 --> Train --> Eval --> Track
    --> Export --> Runtime --> Predict

```

10 Security review

Finding	Severity	Owner	Huong khắc phục
Không có authentication/authorization cho inference va evaluation	Critical	Backend + Solution	thêm API auth, role-based access cho admin flows, enforce tai gateway
CORS đang mở allow_origins=["*"]	High	Backend	khoa origins theo environment
Grafana mac dinh password admin	High	Solution	dua secret ra ngoai Compose va bat buoc rotate
Build args chưa secret DagsHub	High	Solution	chuyen sang runtime secret hoac CI secret mount
CSV upload chưa có file-size/content scanning tot	Medium	Backend	thêm size limit, content-type check, parser hardening

Finding	Severity	Owner	Huong khac phuc
Public evaluation endpoint	High	Backend	dua vao admin surface hoac bo khoi public runtime
Khong co rate limiting	High	Backend + Solution	them control o proxy va application

11 Operations handbook

Tinh huong	Dau hieu	Nguyen nhan	Buoc debug	Khac phuc	Owner
API down	/health fail, UI loading mai	model load fail, crash, path sai	xem backend logs, check MODEL_MODE, verify model files	rollback artifact hoac image	Backend + AI
Model failure	500 tren predict/explain	tokenizer/session mismatch, artifact hong	reproduce API call, inspect ModelError	restore artifact tot cu	AI
Latency spike	p95 tang, Grafana alert	SHAP, ABSA, CPU saturation	tach predict va explain, xem CPU limits	giam heavy path hoac tang tai nguyen	AI + Solution
Monitoring failure	Grafana scrape loi	metrics endpoint hoac config drift	query /metrics, inspect Prometheus targets	sua scrape target hoac data-source	Solution
Batch failure	400/500 khi upload CSV	malformed CSV, thieu cot text	reproduce voi sample CSV, doc parser error	fix input hoac harden parser	Backend
Deployment failure	Compose services unhealthy	config drift, missing volume, build loi	xem docker-compose logs	restore config, rebuild, revert	Solution

12 Technical debt register

Level	Debt Item	Impact	Risk	Owner	Fix Cost	Priority
Critical	Public unauthenticated inference va evaluation	enterprise blocking	security/abuse	Backend	Medium	P0
High	/batch_status la mock	semantics sai	operational confusion	Backend	Medium	P1
High	Language selector khong di den backend	user expectation sai	correctness drift	UI/UX	Low	P1
High	Grafana default credentials va CORS rong	security posture yeu	compromise risk	Solution	Low	P1
Medium	Evaluation state nam trong memory	continuity yeu	stale status	Backend	Low	P2
Medium	SHAP khong co guardrails ro	latency spike	runtime instability	AI	Medium	P2
Low	Frontend test coverage mong	regression detection cham	maintainability	UI/UX	Medium	P3

13 Rui ro to chuc

- **Knowledge silo:** inference internals va artifact promotion dang tap trung vao AI/ML.
- **Single point of failure:** `src/model/baseline.py`, `src/scripts/run_finetuning.py`, va `docker-compose.yml`.
- **Bus factor:** thap cho ONNX export, imbalance logic, observability provisioning.
- **Cross-team dependencies:** thay doi label set can ML, AI, Backend, UI, va tests cung di chuyen.
- **Ownership gaps:** chua co security owner ro rang va release manager chinh thuc.

14 Production readiness review

Area	Score	Status	Ly do
Architecture	74	Amber	layering tot nhung runtime va async semantics chua that
Security	32	Red	thieu auth, CORS rong, default credentials
Testing	76	Amber	Python tests tot, frontend tests mong
Monitoring	71	Amber	co Prometheus/Grafana nhung chua co tracing
Performance	63	Amber	ONNX co loi ich nhung SHAP/ABSA van dat do
Reliability	58	Amber	co health nhung khong co durable queue/state
ML lifecycle	79	Amber	DVC, MLflow, training, eval, export kha day du
Deployment	60	Amber	Compose hop cho env kiem soat, chua hop enterprise
Operations	55	Amber	runbook co the viet duoc nhung tu dong hoa it
Ownership	70	Amber	role kha ro sau handbook nhung van phu thuoc ca nhan
Overall	64	Amber	advanced MVP, chua duyet enterprise production

15 Roadmap

15.1 3 thang

- them authentication, rate limiting, CORS theo environment. Owner: Backend + Solution. Priority: P0.
- thay `/batch_status` bang durable job tracking. Owner: Backend. Priority: P1.
- sua frontend language propagation va dong bo supported-language UX. Owner: UI/UX + Backend. Priority: P1.

15.2 6 thang

- them artifact registry/checksum validation. Owner: AI + ML. Priority: P1.
- them distributed tracing va model SLIs. Owner: Solution. Priority: P2.

- tach admin evaluation khi public inference deployment. Owner: Backend + Solution. Priority: P1.

15.3 12 thang

- chuyen khi Compose-only sang managed deployment co secret va autoscaling. Owner: Solution. Priority: P1.
- formalize model registry, approval gate, rollback policy. Owner: Tech Lead + ML + AI. Priority: P1.
- mo rong multilingual support chi sau khi data, evaluation, UI dong bo that. Owner: ML + AI + UI/UX. Priority: P2.

16 Huong dan onboarding

Moc	Muc tieu onboarding
Day 1	Doc phan lien quan den vai tro: Backend doc <code>src/main.py</code> va <code>contracts/</code> ; AI doc <code>src/model/</code> ; ML doc <code>src/data/</code> va <code>params.yaml</code> ; UI/UX doc <code>app/</code> ; Solution doc Compose va infra.
Day 3	Chay tests lien quan, trace mot request end-to-end, ghi nhan mot discrepancy giua docs va code.
Day 7	Lam mot thay doi nho co review va cap nhat test.
Day 14	Tu minh chay mot workflow day du trong pham vi vai tro.
Day 30	Co the tu nhan mot issue van hanh trong vung so huu va review thay doi cua nguoi khac.

17 CTO review

17.1 Co phe duyét production launch khong?

Khong, neu la production mo hoac enterprise production. Co the phe duyét demo noi bo va pilot rat kiem soat sau khi hardening bao mat co ban.

17.2 Co phe duyét enterprise customers khong?

Khong. Security va operations controls hien tai chua dat muc toi thieu.

17.3 Co phe duyét scaling khong?

Chua. Kien truc co huong scale duoc, nhung deployment va control plane hien tai chua san sang.

17.4 Dieu dang lo nhat la gi?

Nguy hiem nhat la *false confidence*: repo nhin rat giong san pham production, nhung mot so control quan trong van thieu hoac chi la mock.

17.5 Nen dau tu tiep vao dau?

Uu tien hardening nen tang: authentication, rate limiting, artifact governance, va durable batch execution. Sau do moi dau tu manh hon cho multilingual va model operations.

Phu luc discrepancy

- README cho vi du health chi co ["en"], nhung `ModelConfig.supported_languages` la ("en", "vi"). Tac dong: tai lieu noi mot dang, backend support mot dang.
- README mo ta batch theo kieu async job, nhung `/batch_predict` thuc te la synchronous va `/batch_status` chi la mock. Tac dong: client co the tin vao polling semantics khong ton tai.
- Frontend hien nhieu ngon ngu trong `ChatInputComponent`, nhung backend chi support en va vi; service layer con bo qua explicit language. Tac dong: affordance vuot qua nang luc that.
- Compose dung `http://mlflow:5000` noi bo container, host expose 5005, trong khi `params.yaml` default `http://localhost:5000`. Tac dong: assumptions khac nhau giua local CLI va container runtime.